

Accessibility Analysis During Cotranscriptional Folding

Overview

This tool analyzes nucleotide accessibility in secondary structure ensembles during cotranscriptional RNA folding simulations.

Accessibility is defined as how often in the performed simulations a nucleotide is unpaired at a given transcript length.

Input files

The file `dld3.fa` contains a designed RNA sequence:

```
>dld3.fa
UCAGUCUUCGCGUGCGUGUAUCGAUUCGGUUUCAGUUUUUAUUGC
```

The simulation always starts from transcript length = 1 and the initial structure ".".

Input parameters:

The parameter `--t-ext` defines the simulation time per transcript length. To express simulation time in seconds, set the rate constant `k0` to `1e6`.

Simulation setup

Note: Ensure the `fuzzyfold` package is installed.

When working directly from the Git repository, use

```
cargo run --bin ff-accessibility -- [options]
```

instead of calling `ff-accessibility` directly.

To simulate 100 trajectories with an extension time of 0.02 and build the corresponding accessibility profile:

```
cat dld3_lm3.fa | ff-accessibility --macrostates dld3*.ms --k0 1e6 --t-ext
0.02 -n 100
```

or equivalently:

```
ff-accessibility--k0 1e6 --t-ext 0.02 -n 100 < dld3_lm3.fa
```

To view all available options:

```
ff-accessibility --help
```

During execution, the program prints simulation parameters to **STDOUT**, and displays a **progress bar** once all runs are completed.

The accessibility profile is generated automatically as a PNG file. The output file name can be specified via **-output**. If not provided, the default name is accessibility_profile.png.

An example output file from 1000 aggregated simulations of runs with an extension time of 0.02s may look like this:

